

Supporting Material: Doc S1



Uncertainty assessment

Uncertainty ranges for socioeconomic and ecological cycling flows and the resulting input and output cycling rates. See method section for details.

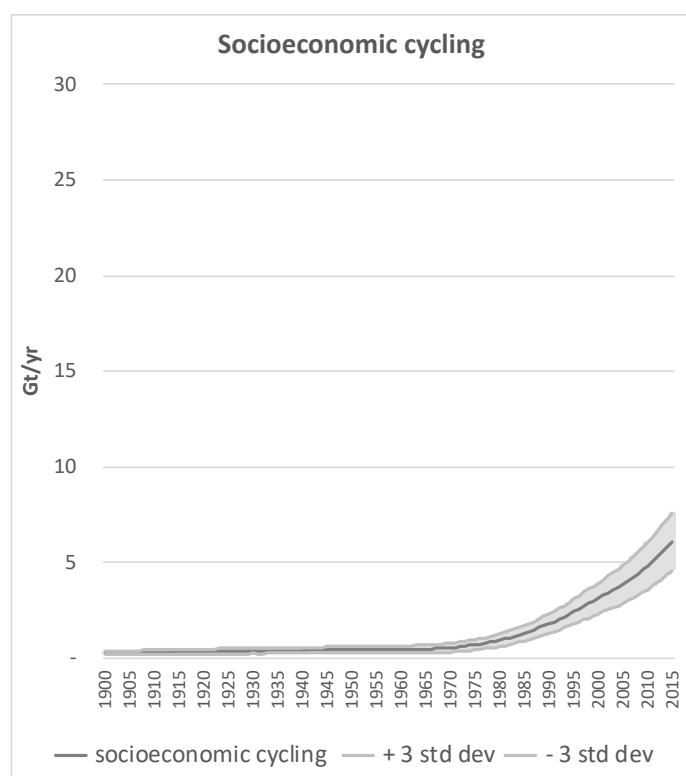


Fig. S1: Range of uncertainty range of secondary materials (socio-economic cycling) for the period 1900 to 2015. The dark grey line represents the values used in this assessment. The light grey lines present 3 standard deviations which cover 99.7% of the 10^3 Monte-Carlo Simulations. Values represented in the grey line were used as maximum and minimum, respectively. Based on Wiedenhofer et al. (2019)¹.

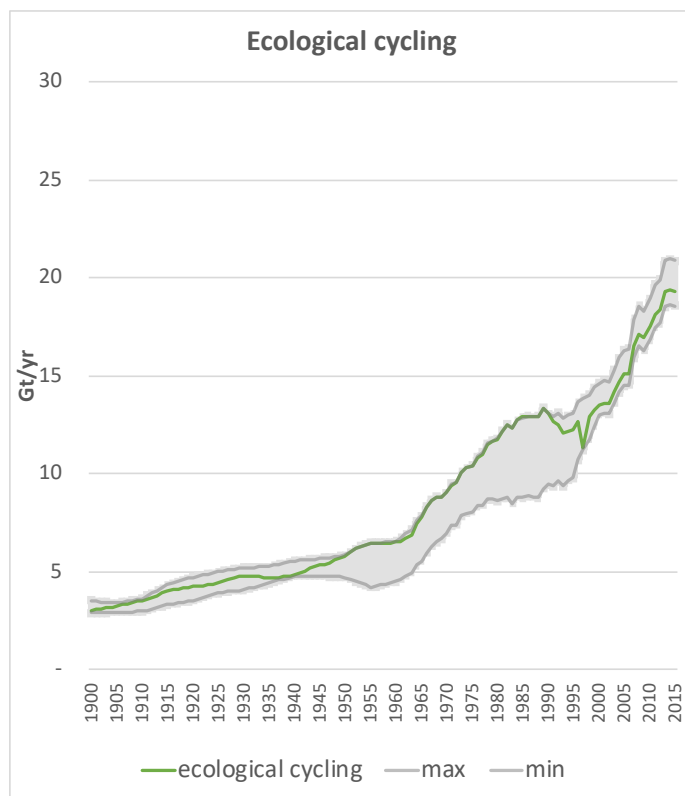


Fig. S2: Range of uncertainty of renewable biomass (ecological cycling) for the period 1900 to 2015 based on different estimates of net-emissions of carbon from land use and land cover change provided in Houghton & Nassikas 2017². The green line represents the values used in this estimate (most recent estimate by Houghton and Nassikas 2017). The upper and lower light grey lines represent the minimum and maximum values of five different estimates. Note that different estimates on net-carbon emissions from land use and land cover change are only available for the period 1900-2000. We extrapolated the uncertainty range keeping the minimum and maximum deviation (in percent) derived constant at the level of the year 2000, i.e. at +8% of renewable biomass for the upper value and -4% for the lower value.

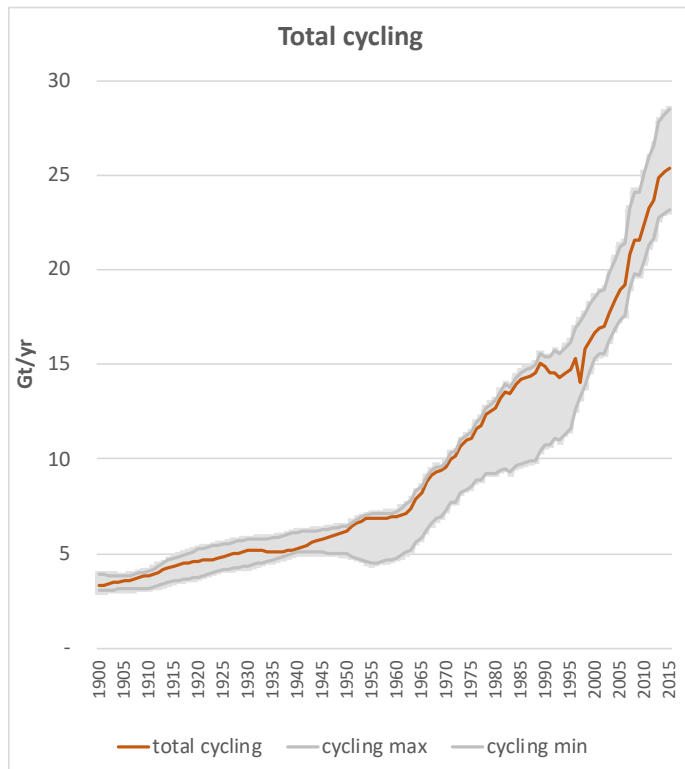


Fig. S3: Range of uncertainty of the total cycling flow (sum of ecological and socioeconomic cycling). The red line represents the values used in this assessment; upper and lower bounds (grey lines) were obtained from adding maximum and minimum values, respectively, of ecological and socio-economic cycling flows shown in Fig. S1 and S2.

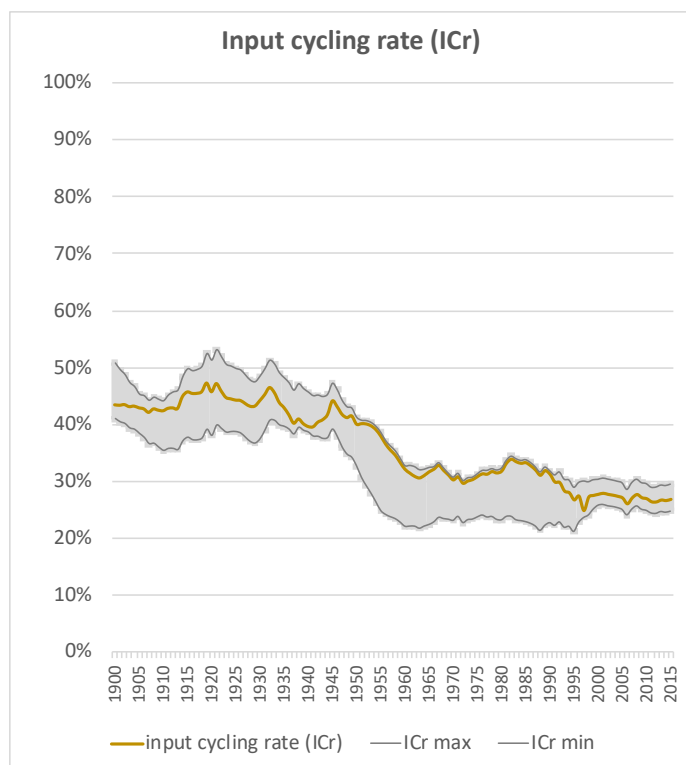


Fig. S4: Range of uncertainty of the input cycling rate (ICr) resulting from the uncertainty of the total cycling flow shown in Fig. S3. The brown line (ICr) is the share of cycling flows in processed materials (as shown in Fig. 3a in the main text), the grey lines represent the calculated minimum (ICr min) and maximum values (ICr max) of the share of cycling flows.

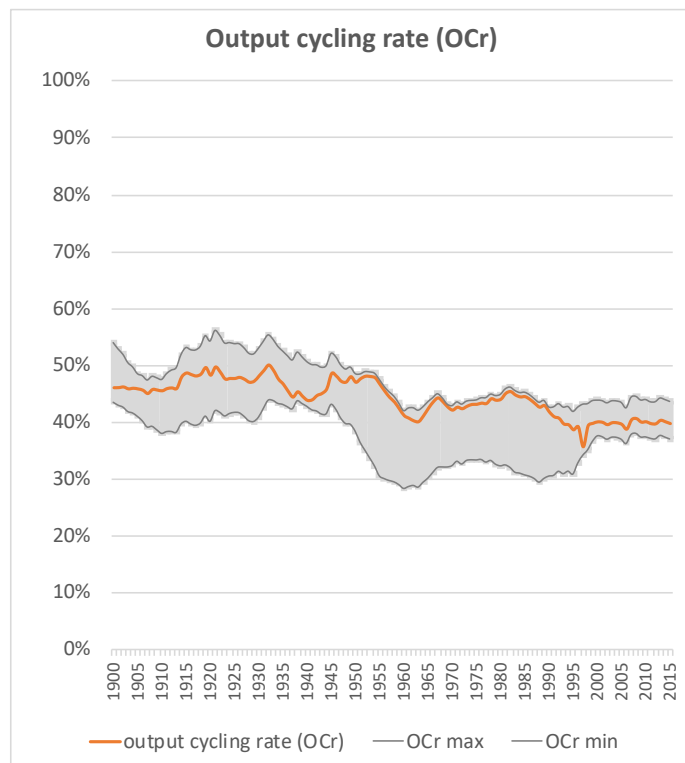


Fig. S5: Range of uncertainty of the output cycling rate (OCr) resulting from the uncertainty of the total cycling flow in Fig. S3. The orange line (OCr) is the share of cycling flows in interim output (as shown in Fig. 3a in the main text), the grey lines represent the calculated minimum (OCr min) and maximum values (OCr max) of the share of cycling flows.

References

1. Wiedenhofer, D., Fishman, T., Lauk, C., Haas, W. & Krausmann, F. Integrating Material Stock Dynamics Into Economy-Wide Material Flow Accounting: Concepts, Modelling, and Global Application for 1900–2050. *Ecological Economics* **156**, 121–133 (2019).
2. Houghton, R. A. & Nassikas, A. A. Global and regional fluxes of carbon from land use and land cover change 1850–2015: Carbon Emissions From Land Use. *Global Biogeochemical Cycles* **31**, 456–472 (2017).